PTM-ECU

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Predictive Thermal Management ECU

Automotive EV Thermal Control | Rev A2 - Architecture, Atomic Requirements, Interfaces & Verification Baseline

Mission profile. The controller coordinates cabin HVAC, battery chilling and powertrain cooling while using route topography, ambient weather and planned charging stops to prepare thermal capacity before high-load events. The product mission includes extended range and a 60-minute optimal fast-charging planning target. That duration is a controlled program parameter whose release definition shall specify the SOC window, usable pack energy, charger capability, ambient and initial pack temperatures, arrival-temperature target and optimization objective.

Engineering baseline. The released architecture retains multi-channel low-side drivers for proportional valves and pumps, 48 V auxiliary-pump operation, an interface boundary to an 800 V e-compressor, CAN FD connectivity to the IVI for route-data ingestion, under-hood IP67 packaging with a pressure-equalization vent, automated PWM-output end-of-line testing and pump-bearing wear estimation from current ripple. Values not yet fixed by the vehicle program remain configuration-controlled and require owner-approved closure evidence before release.

Definition	Controlled product interpretation
Product class	Automotive predictive thermal-management ECU coordinating cabin, battery and powertrain thermal loops.
Safety / assurance	ASIL-B safety intent for prevention of thermal-runaway preconditions. Final ISO 26262 HARA/FSC/TSC allocation and compliance evidence remain program-specific.
Lifecycle mission	Vehicle service life with pump/valve/compressor diagnostics, current-ripple wear indicators, calibration control and change-impact regression.
Primary architecture	Multi-channel low-side actuator drivers; temperature/pressure/current observability; predictive thermal model; CAN FD / IVI route-data interface; 48 V pump and 800 V e-compressor boundaries.
Release philosophy	Runtime faults use bounded degraded states; missing qualification, DVP, manufacturing or supplier evidence blocks the affected variant from release rather than creating a fictitious runtime response.

Future vehicle platforms, pump/valve variants, e-compressor suppliers and predictive-data sources are supported only through explicit interface parameters, controlled calibrations and targeted regression when assumptions change.

PRODUCT GOVERNANCE, REQUIREMENT QUALITY & CHANGE CONTROL

Requirements separate immutable protection limits, variant-controlled actuator characteristics, supplier-dependent device data, vehicle-installation assumptions and calibration values. Fast local protective actions remain independent of cloud, route or infotainment services; predictive information may improve efficiency but shall not increase permissiveness when its validity is unknown.

The global thermal verification envelope includes cold start, hot soak, charging preconditioning, cabin demand, traction heat rejection, degraded cooling, sensor tolerance, source/load transients and beginning/end-of-life actuator behavior. Evidence shall identify hardware revision, software build, calibration set, vehicle/plant model, test equipment and environmental condition.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-GOV-001	Each requirement shall trace to a mission need, hazard/risk control or external interface obligation and to objective verification evidence.	Bidirectional requirement and DVP trace audit.
PTM-GOV-002	Calibration, BOM, software and vehicle-interface variants shall be released as a mutually compatible configuration set.	Configuration-management and variant-compatibility review.
PTM-GOV-003	A change that invalidates thermal, electrical, timing, mechanical or communication assumptions shall reopen only the affected analyses and DVP evidence.	Change-impact and regression-scope review.

Verification, margin and future-change rule

Global limits that depend on pump, valve, compressor, coolant circuit, vehicle platform or regional operating profile remain owned parameters. They shall be closed before release and shall not be converted from planning values into universal product limits.

VEHICLE / THERMAL-SYSTEM BOUNDARY & COMMAND AUTHORITY

The ECU is one element of the vehicle thermal function. It receives vehicle state, battery limits, propulsion thermal demand, cabin requests, ambient information and route/charging context, then commands pumps, valves, fans and the e-compressor interface according to released authority rules. It does not own traction torque, battery contactor authority or charging-station safety.

Boundary failures include conflicting commands, stale vehicle state, incorrect charging forecasts and actuator requests that exceed available electrical or thermal capacity. The state model therefore distinguishes advisory predictive inputs from authoritative local measurements and vehicle safety constraints.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-SYS-001	Every external command and feedback signal shall define source, destination, units, range, update rate, validity, freshness, timeout and fallback behavior.	Vehicle ICD review plus stale/range/timeout fault injection.
PTM-SYS-002	Predictive route or weather data shall not override battery, actuator or hardware protection limits.	HIL authority-conflict and invalid-data test.
PTM-SYS-003	Loss of one thermal-loop input shall transition to the allocated degraded strategy without commanding contradictory actuator states.	Multi-signal dropout and degraded-mode HIL test.

Verification, margin and future-change rule

Changes to vehicle network ownership, battery thermal limits, HVAC authority or compressor control architecture require an interface and safety impact review before prior evidence is reused.

INTEGRATED THERMAL ARCHITECTURE & LOOP DECOMPOSITION

The product coordinates three coupled thermal domains: cabin conditioning, battery chilling/heating and powertrain cooling. Shared coolant paths, valves, heat exchangers or refrigerant interfaces can create cross-coupling, so the architecture must define which controller owns each energy path and how priorities change during fast charging, high propulsion load and cabin comfort demand.

Architecture evidence focuses on energy-flow direction, actuator authority, sensor placement, hydraulic/refrigerant coupling and safe defaults. A command that improves one loop but destabilizes another is treated as a system-level defect rather than an acceptable local optimization.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-ARC-001	The released architecture shall identify thermal-flow paths, shared components, actuator ownership and safe default positions for each operating mode.	Controlled block diagram and thermal/hydraulic ICD review.
PTM-ARC-002	Mode transitions shall prevent simultaneous actuator commands that create prohibited flow short-circuits or loss of required cooling.	State-machine model check and HIL transition test.
PTM-ARC-003	Failure of a shared thermal component shall invoke a vehicle-approved priority strategy for battery, powertrain and cabin loads.	Faulted-component thermal-network simulation and HIL test.

Verification, margin and future-change rule

Coolant topology, shared heat exchanger, refrigerant interface or valve-manifold changes require renewed network correlation and mode-transition verification.

12 V / 48 V POWER INPUT, HOLD-UP & RESET BEHAVIOR

The ECU supports 48 V auxiliary pumps through a defined vehicle low-voltage input architecture that supplies logic and driver rails while separating platform-specific 12 V and 48 V variants. Power sequencing shall prevent unintended PWM during crank, brownout, jump-start, load-dump or supply recovery, and the released feeder assumptions shall be captured in the electrical ICD.

Bias supervision, watchdog state, driver enable and nonvolatile configuration validity are established before actuator outputs become active. Power recovery is bumpless: an ECU reset cannot reopen a valve or restart a pump at an uncontrolled duty solely because retained state is ambiguous.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-PWR-010	Actuator outputs shall remain in their defined non-hazardous reset state until supply, clock, configuration and command-validity conditions are satisfied.	Power sequencing, brownout and reset fault injection.
PTM-PWR-011	Each vehicle power variant shall define normal, abnormal and transient input envelopes used for protection and regulator sizing.	Electrical ICD plus WCCA and transient test.
PTM-PWR-012	Loss and restoration of low-voltage power shall not create an unbounded actuator duty or contradictory valve state.	Crank/brownout/recovery HIL and bench test.

Verification, margin and future-change rule

A change in vehicle supply architecture, feeder impedance, regulator topology or retained-state strategy reopens transient, reset and actuator-default verification.

MULTI-CHANNEL LOW-SIDE PUMP DRIVER ARCHITECTURE

Multi-channel low-side drivers control proportional valves and pumps. Pump channels shall be sized from released steady-state, startup, stall and fault-current envelopes rather than nominal nameplate current. PWM frequency, slew rate, freewheel path and current measurement are coordinated with EMC, acoustic and diagnostic objectives so switching behavior remains compatible with both hydraulic performance and vehicle emissions limits.

Each driver channel needs a defined default-OFF behavior, open-load and short-circuit diagnostics where technically appropriate, thermal protection and a bounded response to driver-supply faults. Current feedback also supports pump-bearing wear estimation through current-ripple features, with diagnostic validity conditioned on PWM operating point, hydraulic load and sampling quality.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-PMP-020	Each pump channel shall remain within released driver SOA for startup, steady-state, blocked-rotor and short-to-supply/ground conditions.	WCCA plus representative motor fault test.
PTM-PMP-021	Pump-driver reset, UVLO or MCU loss shall force the released non-driving state independent of application software.	Reset/UVLO/processor-stall test.
PTM-PMP-022	Pump current measurement used for diagnostics or wear estimation shall retain specified accuracy and bandwidth over the released current and temperature range.	Current-path calibration and dynamic correlation test.

Verification, margin and future-change rule

Pump motor technology, driver IC, freewheel network, harness inductance or PWM strategy changes require channel-specific SOA, EMC and diagnostic regression.

PROPORTIONAL VALVE DRIVER & POSITION / FLOW EFFECT

Proportional thermal valves translate PWM or current commands into hydraulic flow. Their electrical behavior can appear correct while the mechanical valve is stuck, biased or hysteretic, so electrical drive verification is paired with pressure/flow or position evidence where the vehicle architecture provides it.

The ECU limits command slew and applies diagnostics that distinguish an electrical open/short from a mechanically implausible response. Calibration owns the command-to-flow map, hysteresis compensation and default position; these parameters are tied to the released valve family and coolant circuit.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-VLV-030	Valve command-to-effective-flow behavior shall be bounded by a released calibration over supply, temperature and pressure conditions.	Flow-rig characterization and calibration review.
PTM-VLV-031	Electrical valve faults shall be detected and classified without misreporting a healthy channel solely because hydraulic load changes.	Electrical/hydraulic fault matrix test.
PTM-VLV-032	A stale or invalid valve-feedback signal shall not permit unrestricted integral accumulation or repeated destructive actuation.	HIL frozen-feedback and anti-windup test.

Verification, margin and future-change rule
Valve supplier, coil resistance, spool geometry, mounting orientation or coolant-pressure range changes require new command/flow correlation and diagnostic thresholds.

FAN / BLOWER DRIVER COORDINATION & AIRFLOW PLAUSIBILITY

Where the vehicle variant assigns radiator, condenser or cabin blower control to this ECU, airflow actuators are treated as thermal-capacity devices rather than simple PWM loads. Fan speed availability is bounded by supply voltage, back pressure, contamination, temperature and vehicle-speed airflow.

The controller compares requested cooling effect with available speed/current/temperature evidence and avoids assuming that commanded duty equals delivered airflow. A failed fan can require battery or powertrain derating even when other thermal loops remain healthy.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-FAN-040	Released fan channels shall define command limits, startup behavior, diagnostic feedback and thermal derating for the applicable vehicle variant.	Fan bench characterization and vehicle airflow correlation.
PTM-FAN-041	Loss of required airflow feedback shall transition to a conservative cooling-capability estimate rather than a nominal-flow assumption.	Sensor dropout and reduced-airflow HIL test.
PTM-FAN-042	Fan electrical faults shall not propagate through shared supply or ground paths to unrelated critical thermal actuators beyond allocated assumptions.	Power-distribution fault containment test.

Verification, margin and future-change rule
Fan supplier, shroud/radiator geometry, vehicle frontal airflow or driver topology changes trigger airflow and fault-containment revalidation.

800 V E-COMPRESSOR INTERFACE & HV/LV AUTHORITY BOUNDARY

The 800 V e-compressor is treated as an external high-voltage actuator interface rather than assumed to be integrated into the thermal ECU. Compressor torque, speed or cooling-capacity requests therefore require explicit ownership, enable conditions, isolation assumptions, command limits and diagnostic status, with local high-voltage protection remaining inside the compressor or its allocated power-electronics controller.

The thermal ECU shall not create high-voltage authorization on its own. It requests cooling within battery/vehicle permission and reacts to compressor availability, DC-bus state, inverter faults and refrigerant limits supplied by the responsible controller.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-CMP-050	The compressor interface shall define command authority, enable prerequisites, status validity, timeout and non-permissive fallback.	ICD review plus authority/timeout fault injection.
PTM-CMP-051	Loss or invalidity of compressor status shall reduce the assumed cooling capacity before battery or powertrain limits are increased.	HIL frozen/invalid-status test.
PTM-CMP-052	Thermal-control software shall not bypass the compressor controller or vehicle high-voltage safety interlocks.	Interface negative test and safety architecture review.

Verification, margin and future-change rule

Compressor supplier, protocol, HV architecture or refrigerant operating envelope changes require interface and cooling-capacity regression, not a blanket reuse of prior evidence.

ACTUATOR CURRENT, POSITION & PRESSURE OBSERVABILITY

Actuator observability combines electrical current with available position, pressure or flow information to determine whether a command produced the expected physical effect. This prevents a single diagnostic source from confusing a blocked valve, air-locked pump, wiring defect or sensor bias.

Signal qualification includes offset, range, rate-of-change, plausibility and freshness. Diagnostic thresholds are calibrated from real actuator and hydraulic behavior rather than copied from generic counts.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-SNS-060	Each actuator-observability signal shall define accuracy, range, sampling, filtering, plausibility and diagnostic use.	Sensor specification/WCCA and bench correlation.
PTM-SNS-061	Cross-signal plausibility shall distinguish credible hydraulic load changes from electrical or mechanical faults within the diagnostic concept.	Faulted-loop rig and HIL correlation.
PTM-SNS-062	Invalid observability data shall be propagated with quality status and shall not be silently replaced by stale values.	Frozen/stale/out-of-range injection test.

Verification, margin and future-change rule
Sensor technology, scaling, filter bandwidth, mounting or hydraulic topology changes require recalibration of only the affected plausibility paths.

COOLANT TEMPERATURE, PRESSURE & FLOW SENSING

Thermal control depends on temperatures at physically meaningful points and, where available, pressure or flow measurements that reveal pump performance and valve state. Sensor placement and thermal lag are part of the control design because a mathematically accurate sensor can still be unrepresentative of the cell, inverter or coolant hotspot.

The released design documents response time, mounting tolerance, self-heating, bias, open/short behavior and correlation to reference instrumentation. Implausible gradients and frozen signals are detected before they increase allowable power.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-THS-070	Temperature/pressure/flow channels shall meet released accuracy and response requirements at their installed locations over the vehicle environment.	Calibrated thermal/hydraulic rig correlation.
PTM-THS-071	Sensor plausibility shall account for physical thermal lag and expected gradients so valid transients are not classified as faults.	Transient correlation and fault-injection test.
PTM-THS-072	A sensor fault that removes required observability shall invoke the allocated degraded strategy before thermal limits can be relaxed.	HIL sensor-fault timing test.

Verification, margin and future-change rule

Sensor supplier, mounting depth, thermal interface, coolant medium or sampling/filter changes trigger location-specific correlation rather than global retest.

THERMAL STATE ESTIMATION & COOLANT-LOOP MODEL

The predictive ECU needs a thermal state estimate that extends beyond instantaneous sensor readings. The model can represent component thermal mass, coolant transport delay, heat exchanger capacity and actuator response, but its authority is bounded by model validity and measurement correction.

State variables, initial conditions and confidence are reset or constrained after sleep, service, coolant refill or detected sensor faults. Model residuals are monitored so the estimator cannot drift unnoticed while still producing plausible-looking temperatures.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-EST-080	Thermal model states and parameters shall be configuration-controlled to the released vehicle and coolant-loop variant.	Model/configuration review and plant-correlation test.
PTM-EST-081	Estimator residuals shall detect divergence beyond the released model/measurement uncertainty before predicted states increase thermal permissiveness.	Model-mismatch and sensor-bias HIL test.
PTM-EST-082	Initialization after power-up or service shall use bounded states until sufficient valid measurements establish confidence.	Cold-start/service-reset HIL test.

Verification, margin and future-change rule

Heat exchanger, coolant volume, hose routing, pump/valve map or thermal mass changes require model parameter identification and correlation for the affected loop.

BATTERY THERMAL LOOP CONTROL & CELL-PROTECTION PRIORITY

Battery thermal management has priority when cell or pack limits require cooling or heating to avoid safety or charging degradation. The ECU acts on battery-provided temperature and power limits; it does not invent cell chemistry constraints. Predictive preconditioning can prepare the pack but remains subordinate to BMS validity and protection.

Control allocation covers coolant flow, chiller request, compressor availability and shared HVAC resources. If battery data is stale or implausible, the strategy becomes conservative rather than continuing an optimistic predicted temperature trajectory.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-BAT-090	Battery thermal requests shall remain within BMS-authorized limits and shall not override BMS invalidity or thermal protection status.	BMS/thermal ICD review and authority fault test.
PTM-BAT-091	The controller shall define a degraded battery-cooling strategy for stale, conflicting or unavailable pack temperature information.	HIL stale/conflicting battery-data test.
PTM-BAT-092	Battery preconditioning shall be bounded by available cooling/heating capacity and vehicle energy constraints.	Thermal plant/HIL boundary sweep.

Verification, margin and future-change rule

Battery chemistry, pack sensor layout, coolant plate or BMS interface changes reopen only the related thermal model, authority and preconditioning evidence.

POWERTRAIN COOLING LOOP & PROPULSION-HEAT REJECTION

Powertrain cooling coordinates inverter, motor and gearbox thermal demands with available coolant and airflow. Rapid torque transients can create heat after the electrical event, so control uses component temperatures, loss estimates and thermal inertia rather than only present vehicle power.

When shared cooling capacity is constrained, the controller provides a bounded capability indication to the vehicle rather than masking the deficit with aggressive actuator commands. Faults in one branch are prevented from causing unnecessary loss of all thermal capacity.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-PTN-100	Powertrain cooling commands shall respect released component temperature limits and available actuator capacity for the vehicle variant.	Dyno/thermal-rig correlation and limit test.
PTM-PTN-101	Thermal-capacity shortfall shall be reported through the defined vehicle interface before predicted temperature exceeds the allocated control margin.	HIL heat-load ramp and derate-request timing test.
PTM-PTN-102	A branch-level cooling fault shall preserve unaffected loop capability where the hydraulic architecture supports isolation.	Faulted-branch hydraulic test.

Verification, margin and future-change rule
Motor/inverter loss map, coolant routing, radiator capacity or vehicle duty-cycle changes require powertrain heat-rejection model regression.

CABIN HVAC COORDINATION & COMFORT / ENERGY ARBITRATION

Cabin comfort is a major thermal load and can compete with battery chilling or powertrain cooling. The product coordinates comfort requests with safety, energy and range objectives within the released vehicle-program requirements, while preserving battery and powertrain thermal limits when cooling capacity is constrained.

Arbitration is explicit: safety-critical battery or powertrain limits can constrain comfort capacity, while predictive routing may shift conditioning earlier when energy and thermal headroom are available. The transition remains bounded to avoid abrupt airflow or temperature changes caused by stale forecasts.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-HVAC-110	Cabin thermal requests shall be arbitrated against battery and powertrain protection priorities using released mode rules.	Mode-priority HIL matrix.
PTM-HVAC-111	Predictive comfort adjustments shall remain within occupant-command and vehicle-energy authority defined by the program.	Route-prediction and user-command conflict test.
PTM-HVAC-112	Loss of predictive inputs shall return cabin control to a deterministic local strategy without destabilizing shared thermal loops.	Predictive-data dropout test.

Verification, margin and future-change rule
HVAC architecture, heat-pump topology, cabin sensor set or comfort-control ownership changes require arbitration and shared-capacity revalidation.

ROUTE TOPOGRAPHY INPUT & ANTICIPATORY THERMAL CONTROL

Route topography is a predictive input. Grade and expected vehicle load can forecast propulsion heat and energy demand, but route data remains advisory because navigation may reroute or the driver may deviate. The ECU therefore assigns freshness, confidence and prediction horizon to each route-derived estimate and falls back to locally measured thermal states when confidence is inadequate.

Predicted uphill load may pre-cool components or reserve cooling capacity only while the action remains inside local sensor and energy limits. A wrong route shall reduce optimization benefit, not create a safety violation.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-ROUTE-120	Route/topography data used for predictive control shall carry freshness, confidence or validity information and a defined timeout.	Navigation/thermal ICD and stale-data test.
PTM-ROUTE-121	Predictive thermal actions based on route grade shall remain bounded by current local measurements and actuator limits.	Wrong-route and route-change HIL test.
PTM-ROUTE-122	Route loss or divergence shall transition to local-model control without an abrupt actuator or thermal-capacity step.	Navigation dropout/reroute transition test.

Verification, margin and future-change rule
Map provider, navigation protocol, prediction horizon or vehicle-load model changes require only the route-input and anticipatory-control regression.

AMBIENT WEATHER INPUT, SENSOR FUSION & CONFIDENCE

Ambient weather is a predictive input that extends the planning horizon beyond locally measured temperature and humidity. Remote forecast data and onboard ambient sensors are fused with explicit freshness and plausibility handling; a forecast never replaces a local safety-relevant measurement or directly expands actuator authority.

Forecast age, location mismatch and local-sensor disagreement reduce predictive authority. Local hardware protections and measured component temperatures remain dominant.

Engineering closure also checks forecast-to-local-sensor disagreement during rapid weather changes, tunnel entry and solar-load transitions so confidence reduction is physically justified.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-WTH-130	Weather forecast data shall be tagged with age/location validity before it influences predictive thermal scheduling.	Weather-data interface review and stale/location-mismatch test.
PTM-WTH-131	Disagreement between forecast and local ambient sensing shall reduce forecast authority according to released rules.	Sensor/forecast conflict HIL test.
PTM-WTH-132	Loss of weather service shall not inhibit local thermal control or delay protective actions.	Service-outage and concurrent thermal-fault test.

Verification, margin and future-change rule
Weather-service provider, data schema or fusion algorithm changes require interface and confidence-handling verification, not actuator requalification.

CHARGING-STOP PREDICTION & FAST-CHARGE PRECONDITIONING

Planned charging stops and the 60-minute optimal fast-charging planning target inform predictive preconditioning. The duration is not a standalone ECU guarantee; verification depends on the released SOC start/end window, usable pack energy, charger capability, ambient and initial pack temperatures, arrival-temperature target and BMS charging limits. Preconditioning prepares battery and coolant temperatures before arrival while respecting those external constraints.

A planned stop may be cancelled or changed. The controller therefore verifies route confidence and energy availability before investing significant thermal energy, and abandons preconditioning gracefully if the plan becomes invalid.

The preconditioning strategy is evaluated for late arrival, early arrival and cancelled charging events to confirm that stored thermal energy and auxiliary-power use remain bounded.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-CHG-140	Charging-stop data shall define arrival estimate, validity and cancellation behavior before it drives preconditioning.	Route/charging ICD plus cancellation test.
PTM-CHG-141	Battery preconditioning shall remain within BMS, vehicle-energy and actuator-capacity limits for the released variant.	HIL fast-charge preparation boundary sweep.
PTM-CHG-142	Loss or change of the planned charging stop shall unwind predictive actions without violating current thermal constraints.	Reroute/cancel transition test.

Verification, margin and future-change rule

Charging-interface ownership, battery fast-charge limits or route-planner behavior changes require preconditioning logic and interface regression.

STALE / MISSING PREDICTIVE DATA & FAIL-SAFE FALLBACK

Route, weather and charging-plan data are useful but inherently less dependable than local electrical and thermal measurements. The product defines a deterministic fallback that continues safe thermal control when any predictive source is stale, malformed, unavailable or internally inconsistent.

Fallback removes optimization authority first; it does not automatically disable pumps, valves or compressor requests that are still required by local thermal conditions. This distinction keeps communications faults from becoming thermal faults.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-FBK-150	Predictive inputs shall have independent validity and timeout handling so one failed service does not invalidate unrelated local control.	Multi-source stale/dropout HIL test.
PTM-FBK-151	Fallback control shall preserve required local thermal protection while suppressing predictions whose confidence is insufficient.	Concurrent communication/thermal-load test.
PTM-FBK-152	Recovery of predictive data shall be rate-bounded and bumpless to prevent actuator steps caused solely by restored communication.	Service-recovery transition test.

Verification, margin and future-change rule
New cloud, IVI or navigation services require explicit data-quality and recovery rules before they can influence thermal optimization.

CONTROL STATE MACHINE, MODE OWNERSHIP & BUMPLESS TRANSITIONS

The thermal ECU transitions through power-off, initialization, local control, predictive optimization, charging precondition, fast-charge support, degraded and service modes. Each state has explicit actuator authority, accepted inputs and exit criteria.

Transitions are designed around physical energy and fluid dynamics; a software state change cannot instantaneously assume a new coolant flow or refrigerant capacity. Integrator states and actuator commands are transferred or reset to avoid bumps.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-CTL-160	Each control state shall define entry/exit conditions, command authority, accepted data quality and actuator defaults.	State-machine review and transition coverage test.
PTM-CTL-161	Mode transitions shall bound actuator slew and controller state transfer so no unintended thermal or hydraulic step is created.	HIL transition and anti-windup test.
PTM-CTL-162	Conflicting mode requests shall resolve through a deterministic priority rule aligned with the safety concept.	Conflict-injection and state-priority test.

Verification, margin and future-change rule
Any addition of operating modes or authority changes requires state-transition coverage and safety impact analysis.

ISO 26262 ASIL-B SAFETY INTENT & THERMAL HAZARD ALLOCATION

The product carries an ASIL-B safety intent for preventing thermal-runaway preconditions. Final compliance remains dependent on the approved vehicle HARA, FSC, TSC and objective safety evidence. Safety allocation identifies which thermal failures can contribute to battery overheating, propulsion damage or loss of required cooling and assigns independent local protections where predictive-control latency is not credible.

Safety mechanisms are assigned by response time and independence. Fast electrical shutdown remains local to the relevant power electronics, while this ECU owns supervisory detection, actuator coordination, degraded operation and diagnostic reporting within its allocated FTTI.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-SAF-170	Each allocated thermal safety requirement shall trace from HARA/FSC/TSC to mechanism, timing budget and verification evidence.	ISO 26262 traceability and safety review.
PTM-SAF-171	Predictive or network functions shall not be credited as the sole mechanism for a fault requiring faster independent protection.	Safety timing and independence analysis.
PTM-SAF-172	Detected loss of required cooling capability shall transition to the allocated safe/degraded state within the program-defined FTTI.	Fault-injection timing test.

Verification, margin and future-change rule

Vehicle hazard classification, battery safety concept or actuator architecture changes reopen the affected safety allocation before ASIL evidence is reused.

INDEPENDENT HARDWARE PROTECTION & DRIVER SAFE STATES

Application software coordinates thermal actuators, but destructive electrical faults in output drivers must be contained without relying on a slow supervisory task. Driver UVLO, overcurrent/thermal protection and reset defaults are allocated according to the selected hardware.

Protection status is latched or sampled with sufficient fidelity for diagnosis. A faulted driver channel shall not re-enable through repeated software retries unless the safety concept explicitly permits recovery.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-HWP-180	Driver hardware shall force a non-driving or otherwise defined safe electrical state during reset, UVLO and qualifying overcurrent/overtemperature faults.	Driver fault and processor-stall bench test.
PTM-HWP-181	Local hardware protection shall operate within the selected device SOA independent of application-task execution.	Worst-case electrical fault timing test.
PTM-HWP-182	Automatic retry shall be bounded and inhibited for faults classified as potentially destructive or safety relevant.	Repeated-fault/recovery test.

Verification, margin and future-change rule
Driver family, protection architecture or output-energy path changes require channel SOA and protection-timing regression.

THERMAL-RUNAWAY PRECURSOR DETECTION & ESCALATION

The thermal ECU may contribute to prevention of thermal-runaway preconditions by recognizing rising battery temperature, loss of coolant flow, chiller unavailability or other program-defined precursors. It does not replace BMS cell-level safety monitoring and shall not claim cell runaway detection unless allocated by the vehicle safety concept.

Escalation uses confirmed measurements and capability status. Predictive models can provide early warning but cannot suppress a measured hazardous trend.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-TRP-190	Thermal-runaway precursor signals owned by this ECU shall define thresholds, persistence, plausibility and escalation authority in the safety calibration.	Safety calibration review and thermal fault injection.
PTM-TRP-191	A measured hazardous temperature trend shall retain priority over a conflicting benign predictive estimate.	Model-versus-sensor conflict test.
PTM-TRP-192	Loss of required coolant or chiller capability shall be communicated to the BMS/vehicle controller within the allocated timing budget.	End-to-end communication timing test.

Verification, margin and future-change rule
Battery sensor layout, safety ownership or precursor definition changes require renewed HARA/TSC allocation and detection validation.

DIAGNOSTICS, PLAUSIBILITY & FIRST-FAULT PRESERVATION

Diagnostics preserve the first meaningful fault context instead of overwriting it with secondary symptoms. Pump overcurrent, valve mismatch, sensor invalidity, compressor unavailability and network faults are classified with timestamp, operating mode and key thermal variables.

Plausibility rules are designed to avoid nuisance faults during legitimate high-load transients. Diagnostic servicing and logging must not starve time-critical control execution.

Diagnostic correlation retains enough pre-fault and post-fault context to separate primary actuator or sensor failures from secondary thermal consequences during service analysis.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-DIA-200	Safety- or service-relevant faults shall capture first-fault context with configuration identity and sufficient thermal state for root-cause analysis.	Fault-log content and ordering test.
PTM-DIA-201	Diagnostic plausibility shall be validated across normal transients so high thermal demand is not misclassified as actuator failure.	Boundary-condition nuisance-fault test.
PTM-DIA-202	Fault logging and diagnostic communication shall not violate released control-task timing margin.	WCET/load stress test.

Verification, margin and future-change rule
Diagnostic code set, logging schema or service-tool protocol changes require traceability and timing regression, not full thermal requalification.

EMBEDDED SOFTWARE ARCHITECTURE & CONTROL PARTITIONING

The predictive-control concept does not mandate a particular operating system or AUTOSAR profile. The product therefore requires deterministic software behavior while leaving the implementation platform under project configuration control. Software separates actuator I/O, local control, prediction, communications, diagnostics and calibration services, with timing budgets defined for each safety- or performance-relevant path.

The partitioning prevents lower-criticality route or weather processing from starving thermal control. Memory, stack, numeric range and watchdog behavior are bounded for the selected MCU/software platform.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-SW-210	Software architecture shall separate time-critical local thermal control from advisory predictive-data processing with defined interfaces.	Software architecture and freedom-from-interference review.
PTM-SW-211	Runtime memory, stack, CPU and interrupt load shall retain released margin under maximum communication and diagnostic activity.	Target WCET/resource stress measurement.
PTM-SW-212	Numeric overflow, invalid model state or divide-by-zero shall produce bounded outputs and diagnostics rather than uncontrolled PWM.	Unit/SIL numeric-fault testing.

Verification, margin and future-change rule
MCU, compiler, scheduler, OS or major software-partition changes reopen timing, memory and control-stability evidence as applicable.

SCHEDULING, PWM SYNCHRONIZATION & WCET MARGIN

Pump and valve PWM, sensor sampling and control calculations require known timing relationships. Jitter or delayed sampling can distort current-ripple diagnostics, pressure/flow correlation and control stability even when average CPU load appears acceptable.

The timing budget includes interrupt latency, communication bursts, diagnostics and worst-case prediction workload. Watchdog service is tied to successful completion of defined execution milestones.

Timing evidence includes maximum simultaneous PWM updates, CAN FD bursts and diagnostic interrupts so measured reserve reflects the real worst-case execution pattern.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-TIM-220	Control and sensing tasks shall execute at released rates with bounded jitter under maximum defined software and communication load.	WCET/jitter instrumentation on target.
PTM-TIM-221	PWM updates and dependent sensor sampling shall maintain the synchronization required by control and diagnostics.	Timing-trace and phase-correlation test.
PTM-TIM-222	Watchdog servicing shall demonstrate progress through required control milestones rather than simple task liveness.	Task-stall and watchdog fault injection.

Verification, margin and future-change rule

Scheduler, interrupt topology, PWM frequency or sensor-sampling strategy changes require timing and affected control/diagnostic regression.

CAN FD VEHICLE INTERFACE, FRESHNESS & BUS-OFF RECOVERY

CAN FD carries battery limits, thermal requests, actuator status and route-data handoff. Every thermal-relevant message requires explicit authority, range, freshness and timeout behavior because a syntactically valid but stale frame can be more hazardous than complete bus loss. Network recovery shall not create abrupt actuator steps or restore stale predictive authority.

Bus-off, overload and recovery are bounded so communication restart does not instantly restore permissive control from obsolete data.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-CAN-230	Each thermal-relevant CAN FD signal shall define source, range, update rate, freshness, timeout and fallback behavior.	DBC/ICD review plus protocol fault injection.
PTM-CAN-231	Bus-off or prolonged message loss shall transition affected functions to their defined local/degraded behavior without losing independent actuator protection.	CAN fault and recovery HIL test.
PTM-CAN-232	Recovered messages shall satisfy freshness and plausibility criteria before they restore predictive or permissive authority.	Stale-after-recovery negative test.

Verification, margin and future-change rule

Network architecture, gateway ownership, signal encoding or bus load changes require communication timing and data-authority regression.

IVI ROUTE-DATA INTERFACE & TRUST BOUNDARY

CAN FD connectivity to the IVI is used for route-data ingestion. The IVI is a data provider, not a thermal safety authority. Route content is validated for syntax, freshness and plausibility before it changes predictive scheduling, and loss of IVI connectivity shall leave local thermal protection and measured-state control fully available.

An IVI reset, software update or malicious/erroneous message shall not disable local thermal protection or command actuators outside released limits.

Route-data validation also covers IVI reboot, partial message sets and inconsistent destination changes, preventing transient infotainment behavior from appearing as thermal authority.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-IVI-240	IVI route inputs shall be treated as advisory and shall not directly command pumps, valves, fans or compressor actuation.	Interface architecture review and command-injection test.
PTM-IVI-241	Malformed, stale or implausible IVI data shall be rejected without blocking local thermal control.	Negative protocol/HIL testing.
PTM-IVI-242	IVI communication recovery shall re-enter predictive operation only after data freshness and route consistency are re-established.	IVI reboot/recovery test.

Verification, margin and future-change rule
IVI platform, route-data schema or gateway path changes require trust-boundary and data-validation regression.

CYBERSECURITY, SECURE BOOT & CALIBRATION PROTECTION

Networked predictive data operates within the vehicle cybersecurity allocation. The product shall use secure boot, authenticated software, protected calibration and controlled diagnostic access as defined by the program security architecture. Cybersecurity services shall not bypass, postpone or weaken local electrical and thermal protections, and loss of trust shall result in rejection of untrusted data rather than loss of basic cooling control.

Cybersecurity mechanisms shall fail non-permissively for untrusted configuration while remaining independent of immediate local driver protections. A loss of backend or credential service cannot delay thermal protection.

Protected calibration includes actuator maps, derating thresholds and model coefficients because unauthorized changes to these values could alter physical thermal behavior without changing application code.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-CYB-250	Only authenticated released software and calibration shall execute with actuator authority in production-enabled state.	Secure-boot/load and modified-image negative test.
PTM-CYB-251	Maintenance roles shall prevent unauthorized changes to thermal limits, actuator maps or safety-relevant diagnostics.	Role/authorization penetration test.
PTM-CYB-252	Loss of security services shall reject untrusted updates or commands without inhibiting local thermal protection timing.	Concurrent cyber/service-loss and thermal-fault test.

Verification, margin and future-change rule

Trust anchor, bootloader, diagnostic protocol or calibration-signing changes require targeted cybersecurity and configuration-integrity regression.

COMMAND AUTHORIZATION, DATA QUALITY & NON-PERMISSIVE FALLBACK

Multiple controllers may request thermal actions, so the ECU qualifies every request by ownership, operating mode and data quality before increasing capability. The design distinguishes absence of demand from invalid demand; an invalid message cannot be interpreted as permission to stop necessary cooling.

Fallback behavior is function-specific: battery cooling may remain active from local measurements while route optimization is removed. This avoids a single generic fail-safe state that would be technically incorrect across all loops.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-AUT-260	Each command class shall define authorized senders and operating modes before it can influence actuator demand.	Authority matrix and spoofed-source test.
PTM-AUT-261	Invalid or stale commands shall invoke the function-specific fallback and shall not increase thermal permissiveness.	Stale/invalid authority fault injection.
PTM-AUT-262	Conflicting valid commands shall resolve through a deterministic priority rule with diagnostic visibility.	Command-conflict HIL test.

Verification, margin and future-change rule
Changes to vehicle-domain ownership or gateway routing require revalidation of the command-authority matrix and affected fallbacks.

48 V AUXILIARY PUMP POWER, INRUSH & ENERGY MANAGEMENT

The ECU controls 48 V auxiliary-pump demand with awareness of available low-voltage energy, startup inrush and shared vehicle power limits. It shall not assume an unlimited 48 V source. Pump scheduling and derating therefore use released feeder capability, DC/DC availability and vehicle power-management constraints while preserving any safety-critical minimum cooling allocation.

Pump staging can reduce simultaneous inrush, but any sequencing must remain compatible with required thermal response. Low-voltage constraints are communicated to thermal arbitration rather than hidden in a driver saturation.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-48V-270	48 V pump startup and steady demand shall remain within the released vehicle power-network and driver limits.	Power-HIL and pump inrush measurement.
PTM-48V-271	Pump staging or duty limiting shall preserve the thermal response required by the allocated safety and performance modes.	Combined power-limit/thermal-load test.
PTM-48V-272	A 48 V supply fault shall be diagnosed and reflected in available cooling-capacity estimation before demand is increased.	Supply sag/fault injection with capacity monitoring.

Verification, margin and future-change rule

48 V architecture, pump population, feeder impedance or DC/DC source changes reopen only low-voltage power and cooling-capacity evidence.

800 V COMPRESSOR SAFETY BOUNDARY & ENERGY-AVAILABILITY HANDLING

The 800 V e-compressor interface couples thermal capability to high-voltage vehicle state. The ECU must account for HV not-ready, reduced compressor capacity, inverter fault and BMS limitations without attempting to override the responsible high-voltage controller.

Cooling-capacity estimation is updated before the thermal controller assumes a commanded compressor point is available. During HV shutdown, pumps and valves may continue on low voltage according to the released residual-heat strategy.

Residual-heat handling explicitly considers continued coolant circulation after HV loss so stored heat in battery, inverter or refrigerant hardware is not ignored during shutdown.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-HV-280	Compressor-dependent cooling modes shall require valid high-voltage/compressor availability before their capacity is credited.	HV-state/compressor-availability HIL test.
PTM-HV-281	Loss of HV compressor capability shall invoke the released residual-heat and degraded-cooling strategy without unauthorized HV re-enable.	HV shutdown and hot-component test.
PTM-HV-282	Compressor fault/status data shall propagate to thermal arbitration within the allocated timing budget.	End-to-end fault timing measurement.

Verification, margin and future-change rule

HV contactor logic, compressor inverter protocol or residual-heat strategy changes require boundary-specific safety and thermal regression.

THERMAL EFFICIENCY, RANGE OPTIMIZATION & ENERGY ACCOUNTING

The product mission includes extended vehicle range, so predictive thermal control evaluates electrical energy spent on pumps, fans and compressor against avoided battery/powertrain losses and future thermal demand. Optimization remains secondary to safety and required comfort.

Energy accounting uses measured or calibrated actuator power and does not claim savings unless correlated to representative vehicle missions. A prediction error may reduce efficiency but shall not permit thermal limit violation.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-EFF-290	Optimization shall respect all active thermal and safety constraints before minimizing electrical auxiliary energy.	Optimization constraint test over representative missions.
PTM-EFF-291	Energy-benefit claims shall be supported by configuration-controlled vehicle or plant correlation against a defined baseline strategy.	A/B mission-cycle energy test.
PTM-EFF-292	Loss of confidence in the predictive model shall reduce optimization authority while preserving deterministic local control.	Model-confidence degradation test.

Verification, margin and future-change rule
Vehicle mission profile, actuator efficiency map or baseline strategy changes require new energy-benefit correlation, not safety requalification by default.

FAST-CHARGE THERMAL CAPACITY & PRECONDITIONING DVP

Fast charging can create high battery heat load while the vehicle is stationary and cabin demands may remain active. The thermal controller shall arbitrate available cooling against BMS and vehicle limits using the released charge-plan inputs. The 60-minute fast-charging duration remains a mission-level planning parameter, not a universal thermal guarantee, until the program fixes the SOC window, charger power, pack energy and environmental conditions used for verification.

Verification covers preconditioning, arrival temperature, charger start, sustained heat rejection, cooling-capacity limits and recovery after charging.

The fast-charge test matrix includes repeated charging sessions and hot ambient operation to expose capacity limitations that may not appear in a single nominal preconditioning event.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-FC-300	Fast-charge thermal control shall operate within BMS-provided battery limits and released cooling-capacity constraints.	Battery/thermal HIL plus thermal-rig test.
PTM-FC-301	Preconditioning shall not consume thermal or electrical headroom needed for higher-priority protection functions.	Competing-load boundary test.
PTM-FC-302	Loss of cooling capacity during fast charge shall be reported early enough for the vehicle/BMS to execute its allocated power reduction.	Faulted-cooling charge simulation and timing test.

Verification, margin and future-change rule

Battery cell, charging curve, chiller, compressor or coolant-loop changes require fast-charge capacity correlation for the affected variant.

HOT-SOAK RESTART & HEAT-SOAKED ACTUATOR CAPABILITY

After key-off or charging, under-hood components can remain hotter than coolant or ambient sensors suggest. Restart may therefore begin with reduced driver, pump, valve or compressor capability. The ECU uses actual sensor validity and retained thermal state conservatively until sufficient fresh data are available.

A hot reset shall not command full actuator duty merely because a prediction state was lost. Thermal interface aging and restricted airflow are included in end-of-life margin.

Restart verification records driver junction estimate, coolant temperature, enclosure temperature and actuator availability so hot-soak decisions are based on correlated physical evidence.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-HOT-310	Hot-soak restart shall establish valid thermal and actuator-capability state before unrestricted predictive operation.	Hot-soak power-cycle and restart test.
PTM-HOT-311	Driver and actuator duty shall remain within released thermal limits during heat-soaked restart and degraded airflow.	Thermal chamber/load test.
PTM-HOT-312	Invalid retained thermal state shall default to a conservative initialization rather than optimistic beginning-of-life temperature.	NVM invalidation/restart test.

Verification, margin and future-change rule
Enclosure thermal path, driver package, actuator placement or soak mission changes require hot-restart thermal correlation.

COLD-START, VISCOSITY & ACTUATOR AVAILABILITY

Low ambient temperature changes coolant viscosity, pump current, valve response and heat-pump/compressor performance. Cold-start control therefore distinguishes a mechanically slow but healthy actuator from an electrical fault and limits aggressive commands that could exceed driver or pump capability.

Predictive heating may prepare the battery or cabin when energy is available, but the safety strategy remains valid from a fully cold vehicle start without network services.

Cold-start characterization also checks supply-voltage sensitivity and actuator breakaway behavior, since high current alone is not sufficient evidence of a blocked pump or valve.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-COLD-320	Cold-start actuator diagnostics shall account for released viscosity and response-time effects before declaring a fault.	Cold-chamber hydraulic/actuator test.
PTM-COLD-321	Pump and valve commands shall remain within current and mechanical limits across the released low-temperature envelope.	Cold-start current/response characterization.
PTM-COLD-322	Local thermal control shall initialize safely at cold start without dependence on route, weather or cloud availability.	Offline cold-start HIL/vehicle test.

Verification, margin and future-change rule
Coolant formulation, pump/valve supplier or low-temperature vehicle requirement changes trigger cold-start-specific regression.

DERATING, DEGRADED MODES & AVAILABLE-CAPACITY REPORTING

Thermal degradation is expressed as available cooling/heating capacity rather than a single generic fault state. Depending on which pump, valve, sensor or compressor path is lost, the vehicle may retain partial battery, powertrain or cabin capability.

The ECU reports bounded available capacity and reason codes so upstream controllers can reduce charging, propulsion or comfort demand without guessing. Recovery requires confirmed capability, not only cleared diagnostic bits.

Residual capability is recalculated whenever a relevant fault changes flow path, heat-exchanger availability or compressor capacity, ensuring upstream derating requests remain physically meaningful.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-DEG-330	Each credible thermal actuator or sensor loss shall map to a defined residual-capability and degraded-mode response.	Fault matrix and HIL degraded-mode verification.
PTM-DEG-331	Available thermal capacity reported to other controllers shall not exceed capability supported by valid sensors and healthy actuators.	Capability-estimation fault test.
PTM-DEG-332	Recovery from degraded mode shall require validated actuator/sensor function and bounded command re-entry.	Fault-clear/recovery test.

Verification, margin and future-change rule

New actuator redundancy, loop isolation or vehicle derating strategy changes require residual-capability and recovery matrix updates.

UNDER-HOOD THERMAL ENVIRONMENT & ECU SELF-DERATING

The enclosure is installed under hood, so electronics experience ambient heat, conducted heat, vehicle soak and self-heating from multi-channel drivers. The design models PCB hotspots, connector current, enclosure-to-air/structure paths and venting effects.

ECU self-temperature shall not be mistaken for coolant temperature. When electronics thermal margin is reduced, output capability is derated according to driver SOA while preserving the highest-priority thermal functions where possible.

Thermal mapping includes worst simultaneous channel loading and realistic harness current so localized PCB or connector heating is not hidden by average enclosure temperature.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-ENV-340	Electronics hotspot temperatures shall remain within released component limits across the vehicle thermal mission and channel loading.	Thermal mapping and WCCA correlation.
PTM-ENV-341	ECU self-derating shall be based on validated temperature observability and shall preserve deterministic channel priorities.	Thermal chamber multi-channel load test.
PTM-ENV-342	A temperature-sensor fault within the ECU shall not silently assume full driver capability.	On-board thermal-sensor fault injection.

Verification, margin and future-change rule
PCB layout, enclosure material, mounting interface, driver population or under-hood location changes require electronics thermal delta analysis.

IP67 ENCLOSURE, VENTING & CONDENSATION CONTROL

The under-hood enclosure shall achieve IP67 while using a qualified hydrophobic/ePTFE pressure-equalization vent that maintains ingress protection and thermal breathing over temperature cycling, splash, wash and contamination. A specific supplier brand, including GORE-TEX where selected, is controlled in the BOM/AVL rather than mandated by this product requirement unless the released program explicitly freezes that source. Vent placement and membrane characteristics remain part of the pressure-response and sealing validation.

Condensation risk is addressed through sealing, material compatibility and temperature/decompression cycles appropriate to the vehicle program.

Ingress qualification uses production-representative vent installation and sealing process controls because membrane handling or gasket compression variation can dominate real field performance.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-MECH-350	The released enclosure and vent assembly shall meet the program IP67 ingress requirement after relevant environmental and mechanical exposures.	Ingress test on conditioned samples.
PTM-MECH-351	Pressure-equalization strategy shall not compromise sealing, connector integrity or electronics thermal performance.	Thermal/pressure cycling and ingress correlation.
PTM-MECH-352	Vent, gasket and enclosure materials shall remain compatible with under-hood fluids and service cleaning specified by the program.	Material/fluid compatibility qualification.

Verification, margin and future-change rule

Vent supplier, membrane area, enclosure geometry, gasket or mounting orientation changes require targeted ingress/pressure delta verification.

EMC EMISSIONS, IMMUNITY & PWM-DRIVER INTERACTION

Multiple PWM pumps and valves can create conducted and radiated emissions, while under-hood transients and RF exposure can corrupt sensor readings or driver control. EMC design covers supply filtering, grounding, harness return paths, PWM edge rate and sensor common-mode robustness.

Immunity testing monitors physical outputs, not only ECU resets, because an unintended valve or pump pulse can be consequential even if software later recovers.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-EMC-360	The ECU shall meet the vehicle-program EMC emission and immunity limits for the released harness, load and enclosure configuration.	Program EMC test with representative loads.
PTM-EMC-361	EMC susceptibility shall not produce unintended actuator command outside the allocated safety/degraded behavior.	Immunity test with PWM and thermal-state monitoring.
PTM-EMC-362	PWM edge-rate/filter calibration shall preserve both EMC margin and required actuator-control response.	Bench DPT-like edge/EMI correlation and actuator test.

Verification, margin and future-change rule
Harness, PWM frequency, driver IC, filter network or enclosure grounding changes trigger targeted EMC and control-response regression.

CONNECTORS, HARNESS, GROUNDING & SIGNAL-INTEGRITY BOUNDARY

Actuator current returns, sensor grounds and CAN FD signaling share an under-hood harness environment. The interface design controls pin current, contact resistance, ground offset, shield/return strategy and connector sealing so electrical variation does not masquerade as thermal change.

Harness voltage drop and connector heating are included in pump/valve capability and diagnostics. Safety-relevant signals define acceptable ground/reference offset.

Connector thermal rise and ground offset are correlated under combined actuator load, preventing power-return resistance from being misdiagnosed as a sensor or control problem.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-ICD-370	Each connector pin and harness circuit shall define current, voltage, grounding, sealing and diagnostic assumptions for the released variant.	Electrical/mechanical ICD review.
PTM-ICD-371	Harness and contact-resistance tolerance shall be included in actuator voltage/current capability and diagnostic thresholds.	WCCA plus hot-contact/load test.
PTM-ICD-372	CAN FD and low-level sensor interfaces shall remain valid or safely detected under released ground-offset and common-mode conditions.	Ground-offset and communication/sensor test.

Verification, margin and future-change rule

Connector family, pin assignment, harness length/gauge or grounding architecture changes require interface-specific electrical and thermal verification.

CALIBRATION DATA, VARIANT CODING & FIELD-UPDATE CONTROL

Thermal behavior depends on pump maps, valve flow curves, thermal model parameters, derate thresholds and predictive weights. These calibrations are part of the released product configuration and cannot be mixed arbitrarily across vehicle variants.

Calibration updates are authenticated and range checked. A field update that changes a safety-relevant threshold or actuator map requires the same impact assessment as a design change.

Variant coding is checked before actuator enable so a valid but wrong calibration package cannot operate a pump, valve or compressor with incompatible limits.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-CAL-380	Each calibration item shall define owner, units, valid range, variant applicability and verification linkage.	Calibration database and traceability review.
PTM-CAL-381	Out-of-range, corrupt or incompatible calibration shall be rejected before unrestricted actuator control.	Negative-load and variant-mismatch test.
PTM-CAL-382	Field calibration changes affecting safety, control stability or actuator limits shall trigger documented impact analysis and affected regression.	Change-control audit and regression evidence.

Verification, margin and future-change rule
Calibration structure, memory layout or update mechanism changes require integrity, compatibility and recovery verification.

PUMP-BEARING WEAR ESTIMATION FROM CURRENT RIPPLE

Pump-bearing wear estimation uses current-ripple analysis. The diagnostic extracts features only when PWM, load, speed and sampling conditions make the signature meaningful, and it suppresses inference when hydraulic transients or command changes invalidate the model. This avoids turning normal load variation, cavitation events or switching artifacts into false wear predictions.

Wear estimation is advisory/predictive maintenance unless the safety concept explicitly allocates it as a protection input. Thresholds are correlated to representative aged pumps and uncertainty is reported.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-WEAR-390	Current-ripple wear features shall be computed only under released operating windows where sampling and load conditions support interpretation.	Aged-pump bench correlation across load/speed.
PTM-WEAR-391	Wear thresholds shall distinguish hydraulic load variation, electrical faults and bearing degradation with documented confidence.	Fault/aging discrimination test.
PTM-WEAR-392	Invalid or insufficient ripple data shall suppress the wear estimate rather than report a healthy state by default.	Sampling/dropout and off-window test.

Verification, margin and future-change rule

Pump family, PWM frequency, current-sense bandwidth, algorithm or bearing design changes require new wear-signature correlation.

VALVE HEALTH, STICKING & HYSTERESIS DIAGNOSTICS

Valve health diagnostics combine command, electrical current and available flow/pressure response to identify sticking, excessive hysteresis or leakage. The diagnostic is calibrated to the actual valve/manifold because identical duty can produce different flow across pressure conditions.

A suspected mechanical fault is confirmed over appropriate persistence to avoid nuisance flags during transients. If the affected valve controls a safety-relevant cooling path, the residual thermal capability is recalculated.

Valve-health evidence includes seeded friction, partial blockage and leakage cases, allowing the diagnostic to distinguish reduced mechanical authority from a complete electrical failure.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-VHD-400	Valve mechanical-health logic shall use released command/response relationships for the installed valve and hydraulic path.	Valve fault rig and pressure/flow correlation.
PTM-VHD-401	Sticking or excessive hysteresis detection shall include persistence and operating-condition qualification to control nuisance faults.	Transient and seeded-fault validation.
PTM-VHD-402	Confirmed valve degradation shall update residual cooling/heating capability used by thermal arbitration.	HIL valve-fault capability test.

Verification, margin and future-change rule
Valve mechanics, manifold, pressure range or response sensor changes require valve-specific health-correlation updates.

E-COMPRESSOR AVAILABILITY, PERFORMANCE & DIAGNOSTIC CORRELATION

The thermal ECU uses compressor status to estimate refrigerant-side cooling or heating capacity. It distinguishes command rejection, power limitation, overtemperature, HV unavailability and communication loss so the degraded thermal response matches the actual failure.

The ECU does not infer compressor health solely from requested speed. Where refrigerant pressure or temperature data are available, they support plausibility and capacity estimation.

Capacity correlation is repeated across compressor operating regions so a status frame marked healthy is not interpreted as full thermal performance under power or temperature limitation.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-CDI-410	Compressor diagnostic status shall map to defined thermal-capacity consequences rather than a single generic unavailable flag.	Compressor-fault HIL matrix.
PTM-CDI-411	Requested compressor operation shall not be credited as delivered capacity without valid acceptance/status or supporting thermal evidence.	Command/status mismatch test.
PTM-CDI-412	Compressor recovery shall restore capacity gradually according to confirmed status and thermal state.	Fault-clear/recovery test.

Verification, margin and future-change rule

Compressor diagnostic taxonomy, protocol or refrigerant architecture changes require remapping of capacity and recovery behavior.

MANUFACTURING EOL PWM-OUTPUT & LOAD-SIMULATION TEST

Automated end-of-line testing exercises every PWM valve-control output with traceable load or fixture characteristics. The test checks duty cycle, frequency, drive level, diagnostic response and channel mapping so swapped, open or shorted outputs cannot pass production release. Fixture correlation, measurement uncertainty and software/calibration identity are retained with the serialized unit evidence.

EOL evidence is a production release gate, not a runtime ECU state. Fixture uncertainty and golden-unit correlation are controlled so software trim cannot hide an assembly defect.

Production EOL limits are tied to the released driver and load-simulator configuration, with fixture drift monitored so long-term manufacturing variation remains visible.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-EOL-420	Every released PWM actuator output shall be exercised at EOL with objective electrical acceptance limits and channel identity.	Automated EOL fixture record.
PTM-EOL-421	EOL fixture measurement capability shall be demonstrated against calibrated references and controlled golden units.	Gauge capability and correlation study.
PTM-EOL-422	A unit with incomplete or failed critical PWM-output evidence shall not be released as production conforming.	Manufacturing release-gate audit.

Verification, margin and future-change rule

Fixture, test limit, connector adapter, software image or output-driver component changes require EOL correlation before production use.

SENSOR CALIBRATION, OFFSET CONTROL & EOL TRACEABILITY

Temperature, pressure, current and other analog paths may require production calibration or offset verification. Calibration limits are derived from the released measurement error budget and do not compensate for assembly defects beyond approved trim range.

Each stored calibration is linked to serialized hardware and software/configuration identity. Missing calibration evidence blocks release of the affected unit.

Measurement-system uncertainty is included in each EOL pass/fail boundary so calibration acceptance does not consume the engineering margin reserved for field operation.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-EOL-430	Production-calibrated sensor channels shall meet released offset/gain limits with traceable instrumentation.	EOL calibration and metrology record.
PTM-EOL-431	Software trim shall remain within approved calibration bounds and shall not conceal a hardware defect that consumes safety or performance margin.	Golden-unit and out-of-family negative test.
PTM-EOL-432	Calibration data shall be traceable to unit serial number, hardware revision and released software/configuration.	Genealogy/configuration audit.

Verification, margin and future-change rule
Sensor supplier, ADC path, calibration fixture or trim algorithm changes require measurement-system correlation and affected channel revalidation.

PROCESS CAPABILITY, SERIALIZATION & CONFIGURATION GENEALOGY

Manufacturing controls critical characteristics such as driver assembly, connector seating, sealing, vent installation, analog component tolerance and programmed configuration. Process capability is demonstrated for features that consume electrical, thermal or ingress margin.

Serialization ties PCB, enclosure, software, calibration and EOL evidence to the shipped ECU. An incompatible BOM/software/calibration combination is a release blocker.

Process controls identify special characteristics whose variation can affect driver SOA, connector heating, sealing, sensor accuracy or vent performance. Control-plan reactions prevent out-of-family assemblies from being released while preserving full genealogy for containment and corrective action.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-MFG-440	Critical manufacturing characteristics shall have documented inspection or process-capability controls appropriate to their risk.	Control plan, capability and inspection audit.
PTM-MFG-441	Each serialized unit shall retain genealogy linking hardware, software, calibration and EOL results.	Manufacturing traceability audit.
PTM-MFG-442	Configuration rules shall prevent release of incompatible hardware/software/calibration/BOM combinations.	Variant-lock and negative-build test.

Verification, margin and future-change rule
Process, site, supplier or programming-flow changes require manufacturing impact review and requalification of affected critical characteristics.

SYSTEM HIL, THERMAL-RIG & VEHICLE DVP

Verification combines SIL/HIL for state logic and faults, thermal/hydraulic rigs for physical actuator response, and vehicle or representative system testing for coupled-loop performance. Acceptance criteria are fixed before execution and include raw data retention.

The DVP covers normal corners, credible faults, combined stresses and recovery. Incomplete evidence keeps the verification gate open; it is not represented as a runtime derate command.

Correlation between HIL plant models and physical rigs is reviewed periodically so simulation evidence remains representative when actuators, coolant circuits or controller calibrations evolve.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-DVP-450	The DVP shall cover released operating modes, boundary conditions, credible faults and combined thermal/electrical/network stresses.	Configuration-controlled DVP matrix.
PTM-DVP-451	Each critical test shall identify instrumentation calibration, hardware/software/calibration identity and objective pass/fail limits.	DVP report and raw-record audit.
PTM-DVP-452	A variant lacking required critical DVP evidence shall not claim production equivalence or released thermal performance.	Release-gate review.

Verification, margin and future-change rule

Plant model, vehicle platform, actuator family or major control architecture changes require documented similarity analysis before prior DVP is reused.

ENVIRONMENTAL, VIBRATION, INGRESS & SEQUENTIAL QUALIFICATION

Environmental qualification is applied to the released under-hood installation and includes temperature cycling, vibration/shock, electrical transients, fluids, ingress and EMC as required by the vehicle program. Sequential testing is important because sealing or connector damage may only appear after combined exposures.

Missing qualification evidence is a hardware-variant release blocker. Runtime software cannot compensate for an unqualified enclosure or mechanical stack.

Combined-environment sequences are selected to expose interactions such as vibration-induced seal degradation followed by ingress, rather than treating each qualification stress as independent.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-QUAL-460	Environmental qualification categories and severities shall be derived from the actual vehicle installation and program standards.	Qualification plan and installation review.
PTM-QUAL-461	Post-environment electrical, sealing, calibration and actuator-interface performance shall remain within released limits.	Sequential environment plus functional test.
PTM-QUAL-462	A hardware/installation variant with incomplete required environmental evidence shall not be released until the qualification gap is closed.	Qualification release-gate audit.

Verification, margin and future-change rule

Installation zone, enclosure, connector, mounting or vent changes require environmental delta analysis and only the affected regression.

LIFECYCLE, AGING COUNTERS & SERVICE DIAGNOSTICS

Lifecycle monitoring includes pump-bearing wear estimation from current ripple together with actuator-cycle counts, electronics thermal exposure, seal/vent aging, calibration drift and software/configuration history where those factors affect remaining product margin. Service thresholds shall use validated indicators and shall not convert an unqualified trend feature directly into a maintenance or derating decision.

Service data are used to support maintenance and field analysis without silently changing safety limits. Aging thresholds are tied to validated component models and field correlation.

Lifecycle evidence distinguishes reversible service degradation from irreversible aging, allowing maintenance thresholds to be based on validated wear mechanisms rather than accumulated time alone.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-LIFE-470	Lifecycle counters used for maintenance or derating shall define source, units, persistence and rollover behavior.	Long-duration counter and NVM endurance test.
PTM-LIFE-471	Wear or aging models shall be correlated to representative component life data before they drive service recommendations.	Life-model correlation review.
PTM-LIFE-472	Field-data or service-tool updates shall not modify immutable safety limits outside approved configuration control.	Service/update authorization test.

Verification, margin and future-change rule

Mission-profile extension, new aging mechanism or field-derived threshold changes require lifecycle model impact review and affected DVP.

SUPPLIER CHANGE, SECOND SOURCE & OBSOLESCENCE CONTROL

A pump, valve, sensor, driver, vent, connector or processor second source is not production-equivalent merely because nominal specifications match. Similarity review compares the parameters that matter to control, diagnostics, EMC, thermal margin, sealing and safety independence.

Where similarity cannot be demonstrated, affected analyses and tests are repeated before the new source is approved. Missing supplier evidence blocks release rather than forcing an artificial runtime state.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-SUP-480	Second-source approval shall identify all changed characteristics that can affect electrical, thermal, control, mechanical, EMC or diagnostic behavior.	Supplier-change impact matrix.
PTM-SUP-481	Prior verification may be reused only when documented similarity shows changed parameters remain inside validated assumptions.	Similarity and evidence-reuse review.
PTM-SUP-482	A changed component lacking required similarity or regression evidence shall not be approved for production equivalence.	Supplier-release gate audit.

Verification, margin and future-change rule
Obsolescence redesigns are treated as controlled engineering changes with targeted regression proportional to the changed assumptions.

RELEASE EVIDENCE INDEX & ENGINEERING ASSURANCE CLOSURE

Release combines architecture/ICD closure, safety allocation, hardware/software margin, calibrated configuration, DVP, manufacturing capability, environmental qualification and lifecycle evidence. A descriptive paragraph alone does not close a requirement; completion is tied to the specified test, analysis, inspection or configuration record.

Open TBC values have named owners and closure evidence. Waivers for failed or incomplete high-severity mechanisms require approved rationale and cannot silently become production assumptions.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
PTM-REL-490	Release shall require closure of controlled product limits, interfaces, safety/risk allocations and variant definitions.	Architecture/ICD/safety release review.
PTM-REL-491	Critical requirement closure shall reference objective evidence identifying the released hardware, software and calibration configuration.	Requirements-to-evidence audit.
PTM-REL-492	Any unresolved gap affecting safety, thermal capacity, electrical SOA, environmental qualification or production conformity shall block release of the affected variant.	Final engineering assurance gate.

Verification, margin and future-change rule

Future platform, actuator, software, protocol or supplier changes may reuse prior evidence only inside documented similarity bounds. Any change affecting thermal capacity, timing, safety independence, electrical stress, EMC, ingress or lifecycle margin triggers targeted impact analysis and regression before equivalence is claimed.